

PORTFOLIO

/ Tye Cao

01 Bradleys Head Floating Theatre

BDES3027 Architecture Studio 3B

/ Mosman, NSW 2088

02 Infrastructure Re-imagined - Track 19

BDES3026 Architecture Studio 3A

/ Eveleigh, NSW 2015

03 Gibbs Courtyard House

BDES3025 Professional Practice

/ Newtown, NSW 2042



01 BRADLEYS HEAD FLOATING THEATRE

Location	Bradleys Head Rd, Mosman NSW 2088
Main Function	Floating Performing Arts Centre & Public Amphitheatre
Structure	Lightweight Concrete & Steel Framework, Tethering/Anchoring System
Tutor	Emma Justine Anderson
Year	2023
Key Words	Floating Architecture, Sunset Interactivity, Water Ripples Typology
Software	Rhino, Illustrator, Photoshop

Situated on the northern coast of Sydney Harbour at Bradleys Head, the site commands a 220-degree unobstructed panoramic vista of the city skyline. Inspired by daily ferry journeys, the project blurs traditional theatrical boundaries, seamlessly integrating the built form into the maritime landscape.

Rather than constructing a static monument upon the water, the design functions as a dynamic interface with the environment—leveraging the shifting gradients of natural light during the radiant golden hour, and a morphology inspired by water ripples to celebrate the intersection of visual dance, performing arts, and the public domain.

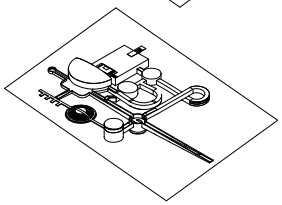
Arrange the modules based on the theater's initial functional circulation.

Based on the concept, reconfigure the layout to orient the main venue towards the west.

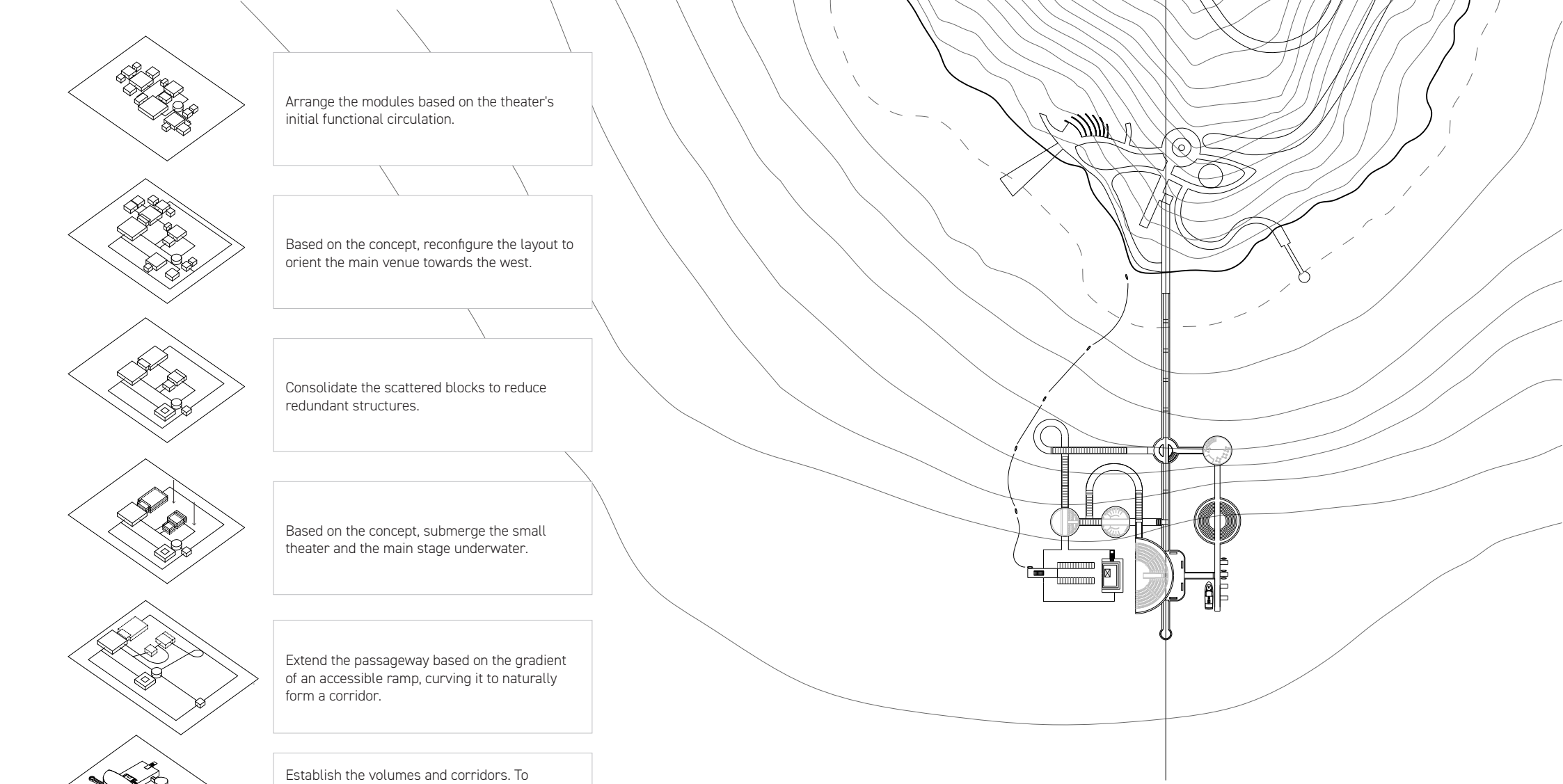
Consolidate the scattered blocks to reduce redundant structures.

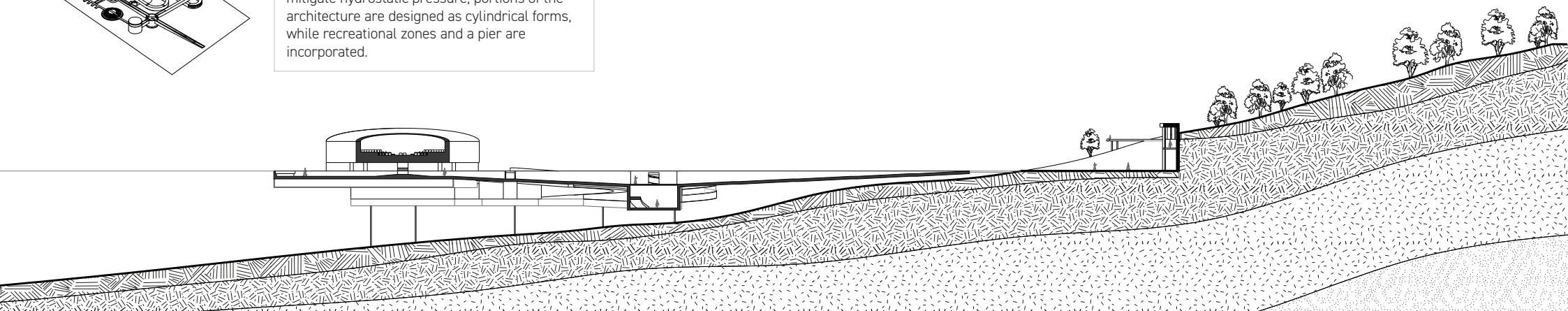
Based on the concept, submerge the small theater and the main stage underwater.

Extend the passageway based on the gradient of an accessible ramp, curving it to naturally form a corridor.



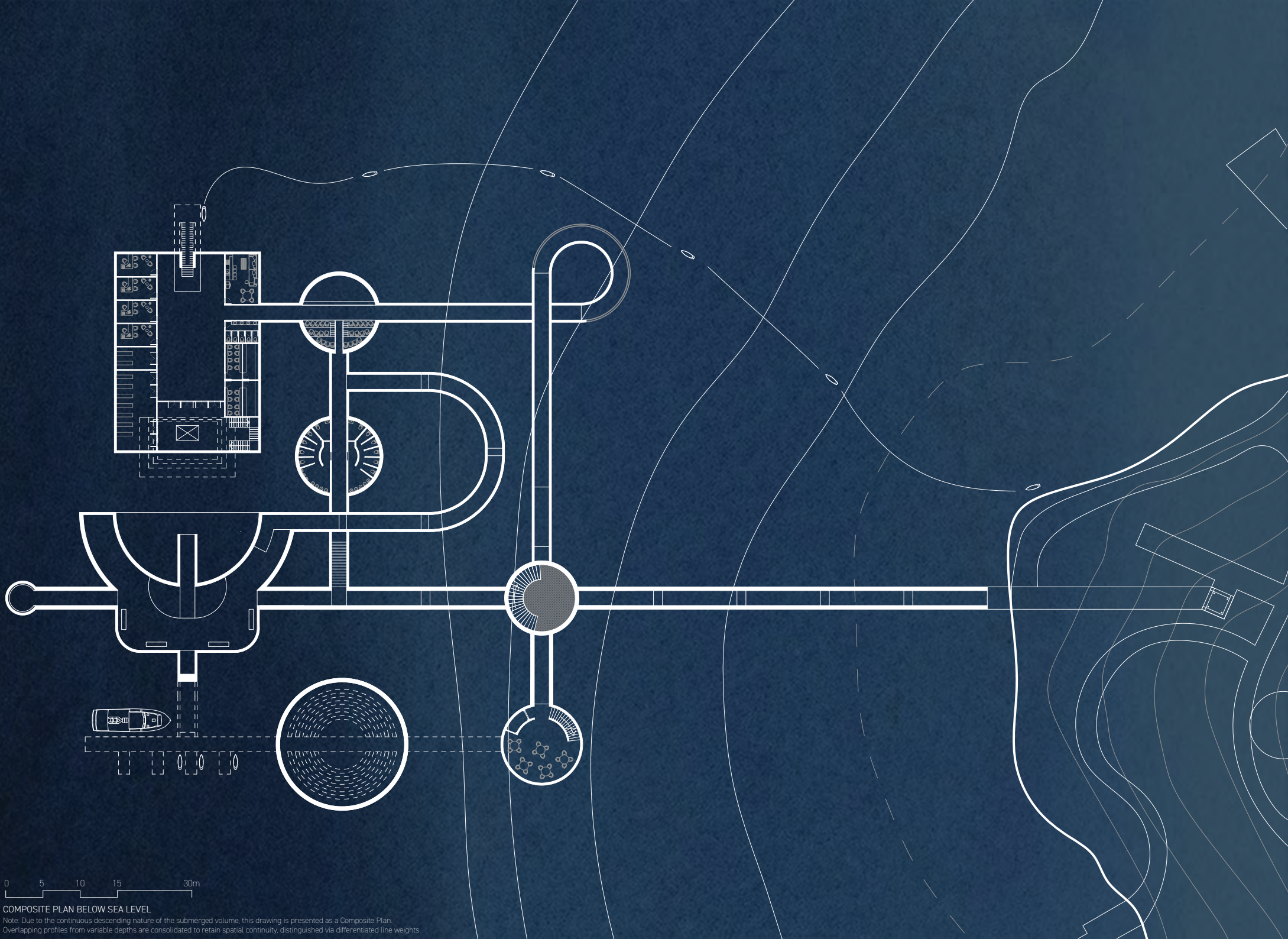
Establish the volumes and corridors. To mitigate hydrostatic pressure, portions of the architecture are designed as cylindrical forms, while recreational zones and a pier are incorporated.





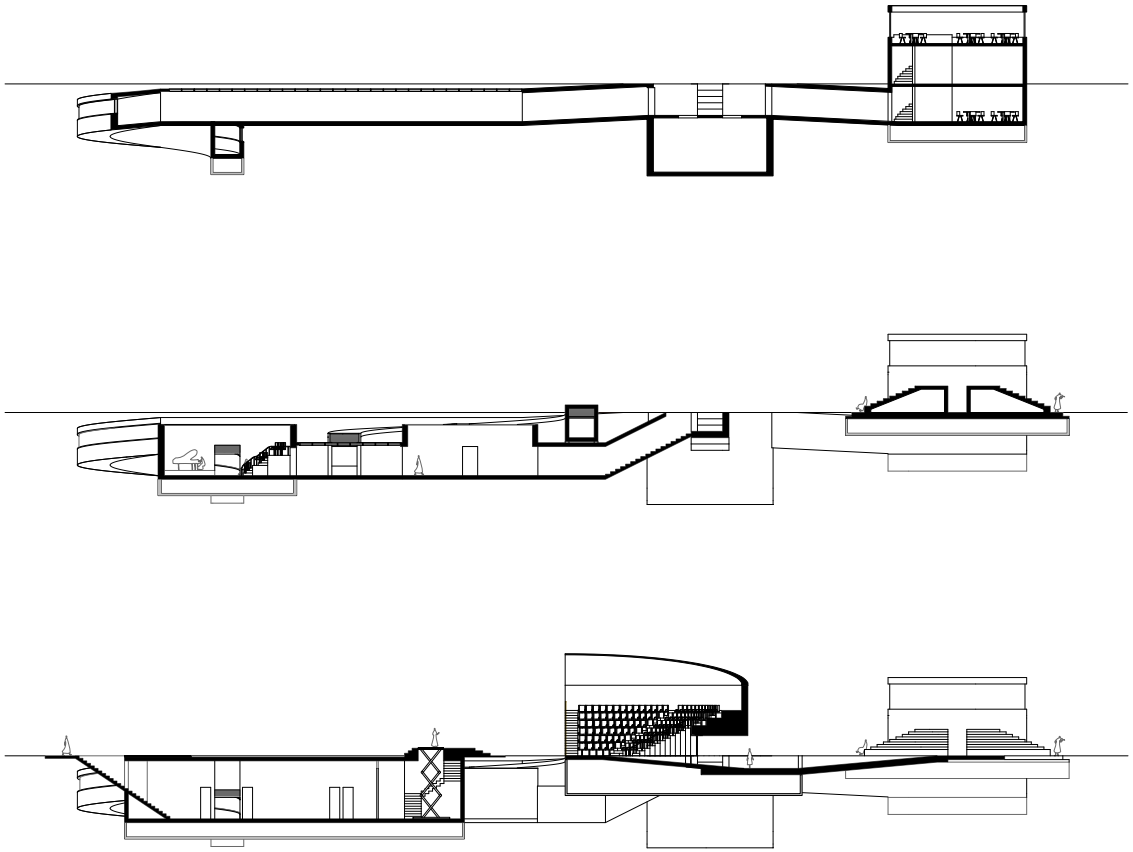


PLAN ABOVE SEA LEVEL

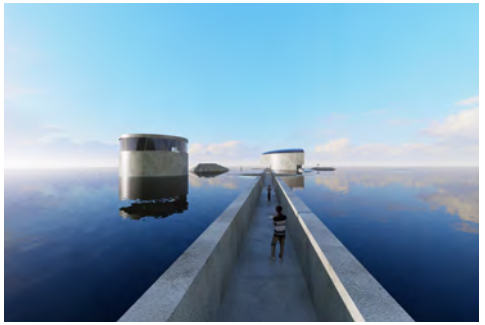


COMPOSITE PLAN BELOW SEA LEVEL

Note: Due to the continuous descending nature of the submerged volume, this drawing is presented as a Composite Plan.
Overlapping profiles from variable depths are consolidated to retain spatial continuity, distinguished via differentiated line weights.





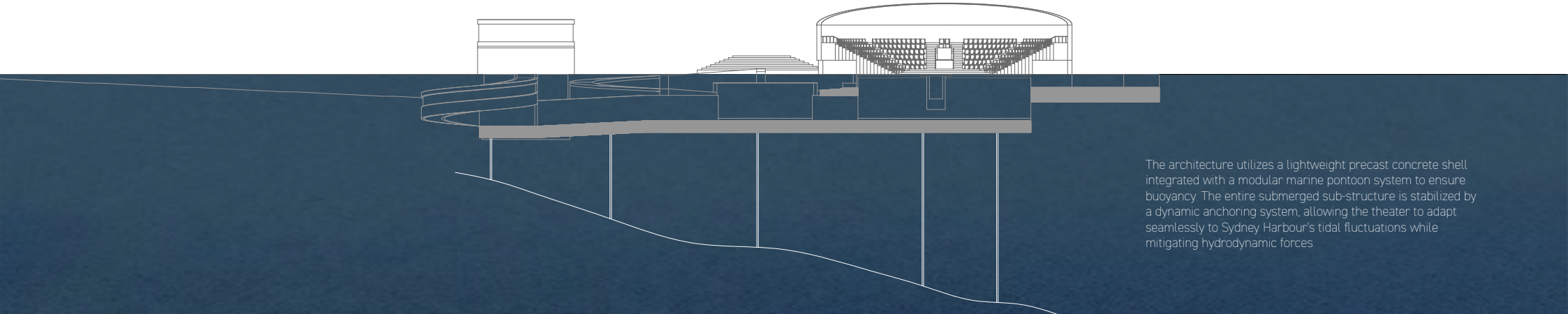


THE ABOVE-WATER THEATER & CIRCULATION

Sunset Framing
The main venue is oriented toward the west to perfectly frame the twilight and the setting sun, ensuring an unobstructed view of the bay's scenic beauty from the central amphitheater.

The Aquatic Veil
To create an immersive performance surface, the backstage roof is submerged precisely 10cm beneath the water surface. This thin aquatic veil renders the backstage structures invisible under daylight, allowing actors to miraculously walk on water during performances.

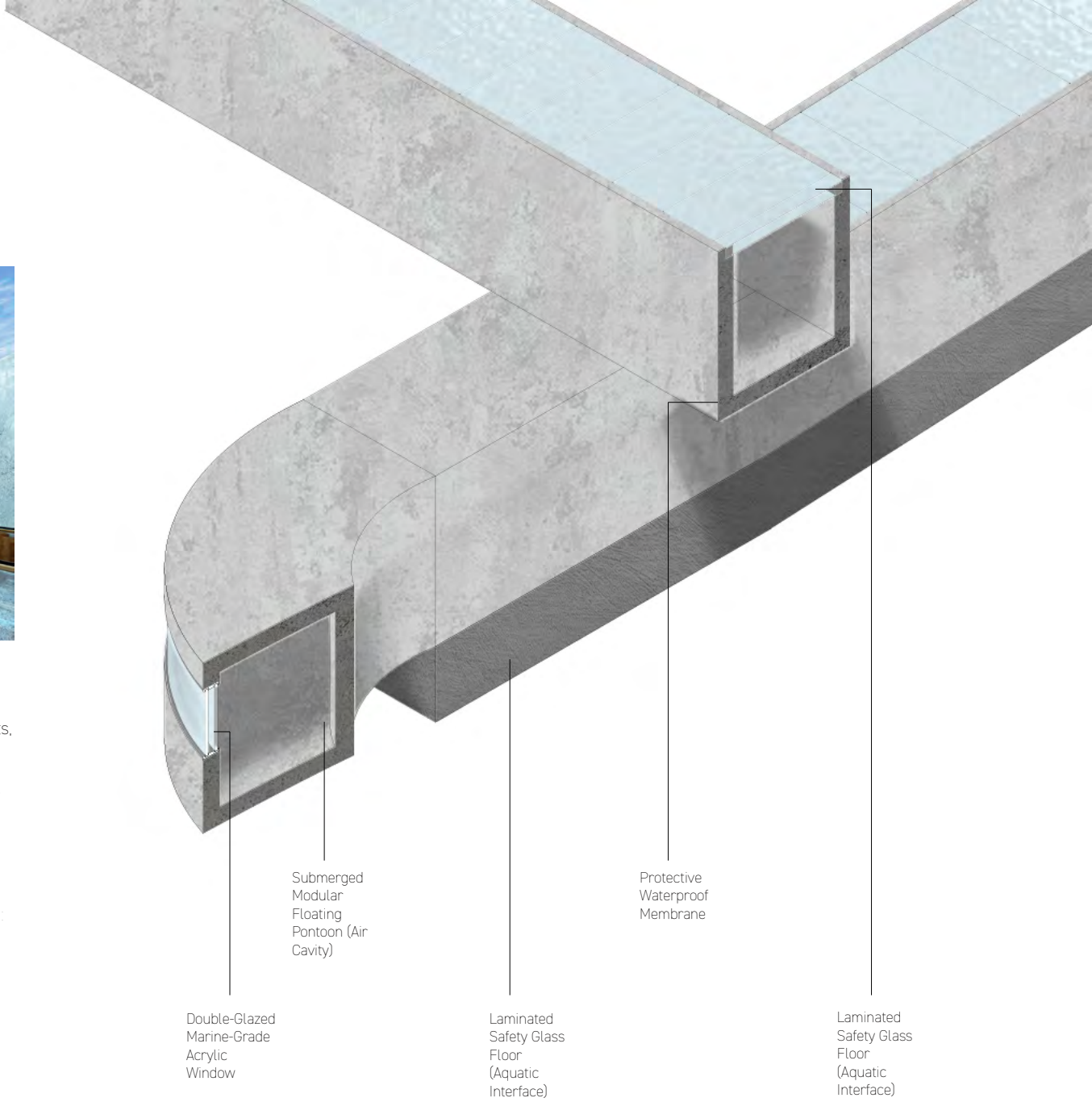
Multimodal Gateway
The circulation operates as a complex infrastructural gateway. An extended minor stage acts as a focal point for boat docking, while the eastern edge opens into an open-air civic plaza, connecting distinct panoramic dining and intimate waterfront experiences.



The architecture utilizes a lightweight precast concrete shell integrated with a modular marine pontoon system to ensure buoyancy. The entire submerged sub-structure is stabilized by a dynamic anchoring system, allowing the theater to adapt seamlessly to Sydney Harbour's tidal fluctuations while mitigating hydrodynamic forces.



Going deeper into the heart of the building, a tranquil underwater stage awaits, where light pierces the surface to cast captivating patterns, and continuous, unpredictable blue water ripples playfully animate the raw concrete interior. Hydrophobic monolithic concrete was selected as the primary material for its superior resistance to underwater environments and its ability to ground the architecture in deep water. To mitigate hydrostatic pressure, the layout consolidates scattered blocks into cylindrical geometric forms under the sea level. The journey to this subterranean sanctuary is defined by an extensively elongated corridor. Beyond fulfilling regulatory compliance for accessible ramp gradients, this prolonged ramp is a deliberate phenomenological device: it stretches the perception of time and deepens the sense of meditative descent. As visitors traverse this shifting spatial progression, the spatial contrast of light and shadow intensifies, culminating in a visual and auditory feast where dance and music converge under a shining, iridescent aquatic dome.



02 Infrastructure Re-imagined - Track 19



Location	North Everleigh, Refern, NSW 2015
Main Function	Clean Energy Research Hub, Community Multipurpose Spaces
Structure	Mass Timber Post-and-Beam Structure
Tutor	Melissa Liando
Year	2023
Key Words	Industrial Heritage, Urban Connectivity
Software	Rhino, Illustrator, Photoshop

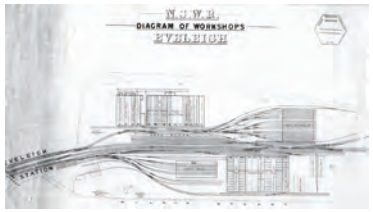


01 / CONTEXT & HERITAGE SYNTHESIS

Site Genesis (1878–1989)
Established in 1878, the Eveleigh Northern Yard served as the heart of the NSW rail network for over a century, dedicated to locomotive manufacturing and carriage maintenance.

Urban Convergence
Today, this distinct industrial corridor interfaces three conflicting urban fabrics: the high-density USYD academic precinct, a dense residential neighborhood, and active transit infrastructure.

Design Response
The design bridges these zones by functioning as a "green lung"—safeguarding solar access for neighbors, managing dense campus pedestrian flows, and weaving historic train culture into contemporary community life.







02 / DESIGN STRATEGY & CONCEPT

Geometric Catalyst
Rather than a static relic, the historic rail turning platform (roundhouse) inspires the building's layout. The architecture adopts its circular radial logic to dynamically converge new volumes around a central node, generating elevated, sweeping walkways..

Material Dialogue
Weathering steel (COR-TEN) is selected as the primary facade language. Its evolving, rust-like patina directly echoes the 19th-century industrial iron textures, establishing a tactile dialogue between the past and the contemporary form.

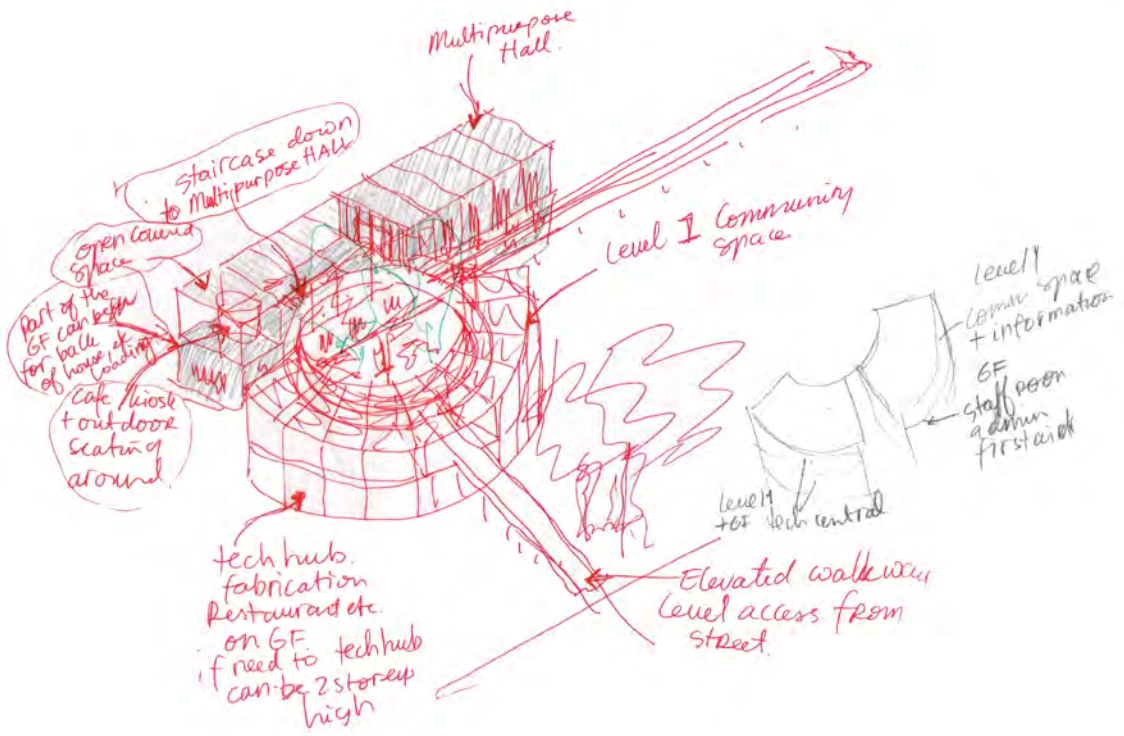
Architectural Vision
The revitalization weaves collective memory into public space through four tactical moves:

- Energy Hub: Showcasing clean energy evolution.
- Track Memorial: Preserving raw rail infrastructure.
- Viewing Platforms: Engaging visually with the active Sydney rail network.
- Elevated Walkways: Utilizing sweeping Corten paths to seamlessly unify disparate program zones.



03 / DESIGN COLLAGE

This composition serves as a material and historical manifesto for the project, utilizing a COR-TEN steel canvas to anchor the narrative in industrial memory. The dynamic expansion of the rail lines symbolizes the historical trajectory of transit innovation. The radial convergence of the four locomotive eras—ranging from early steam power to future clean-energy systems—mirrors the overarching programmatic intent of the tech hub: to act as a living archive and a forward-looking research node for energy systems and urban mobility.

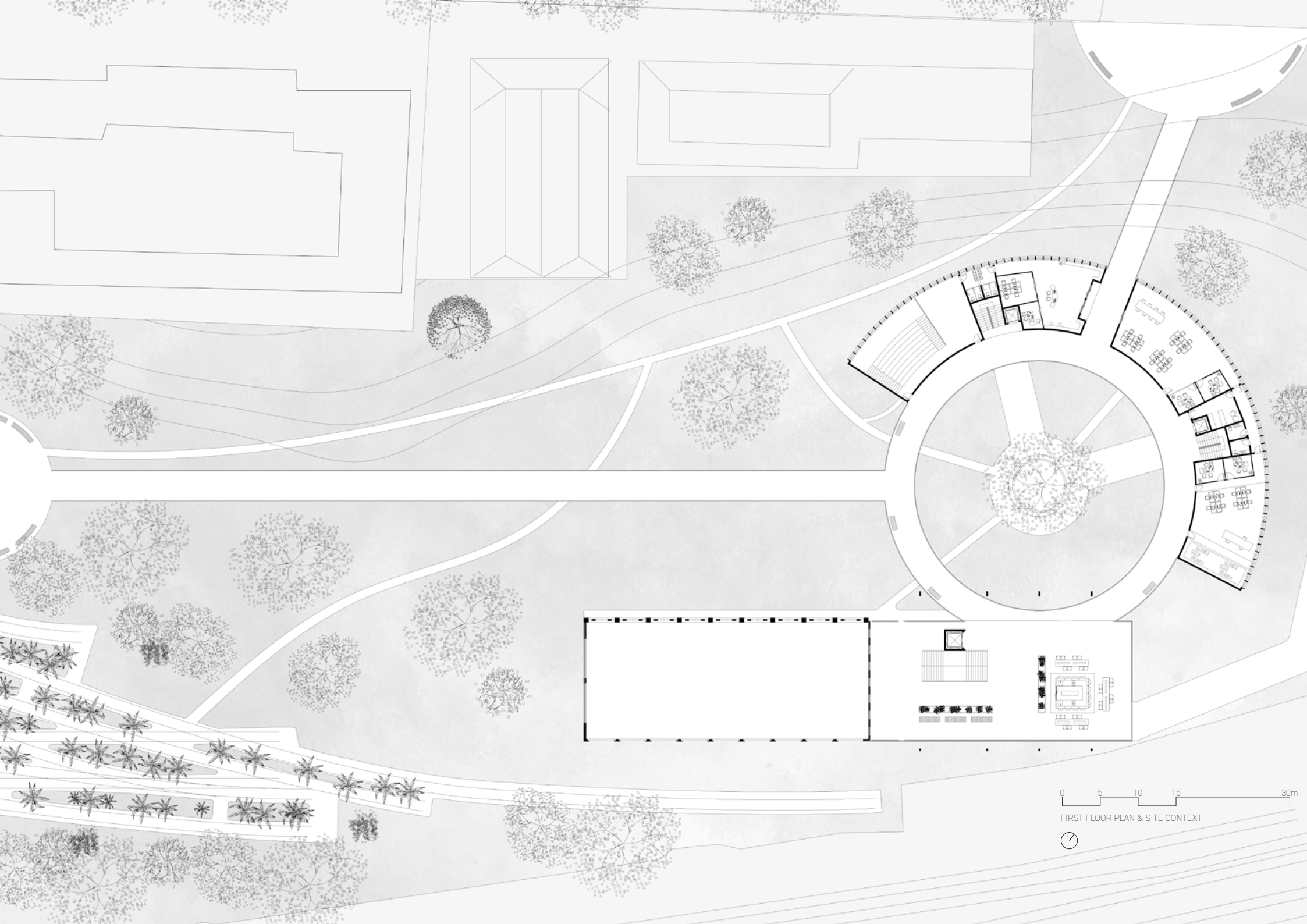


04 / SPATIAL SEQUENCE: THE TANDEM STATION

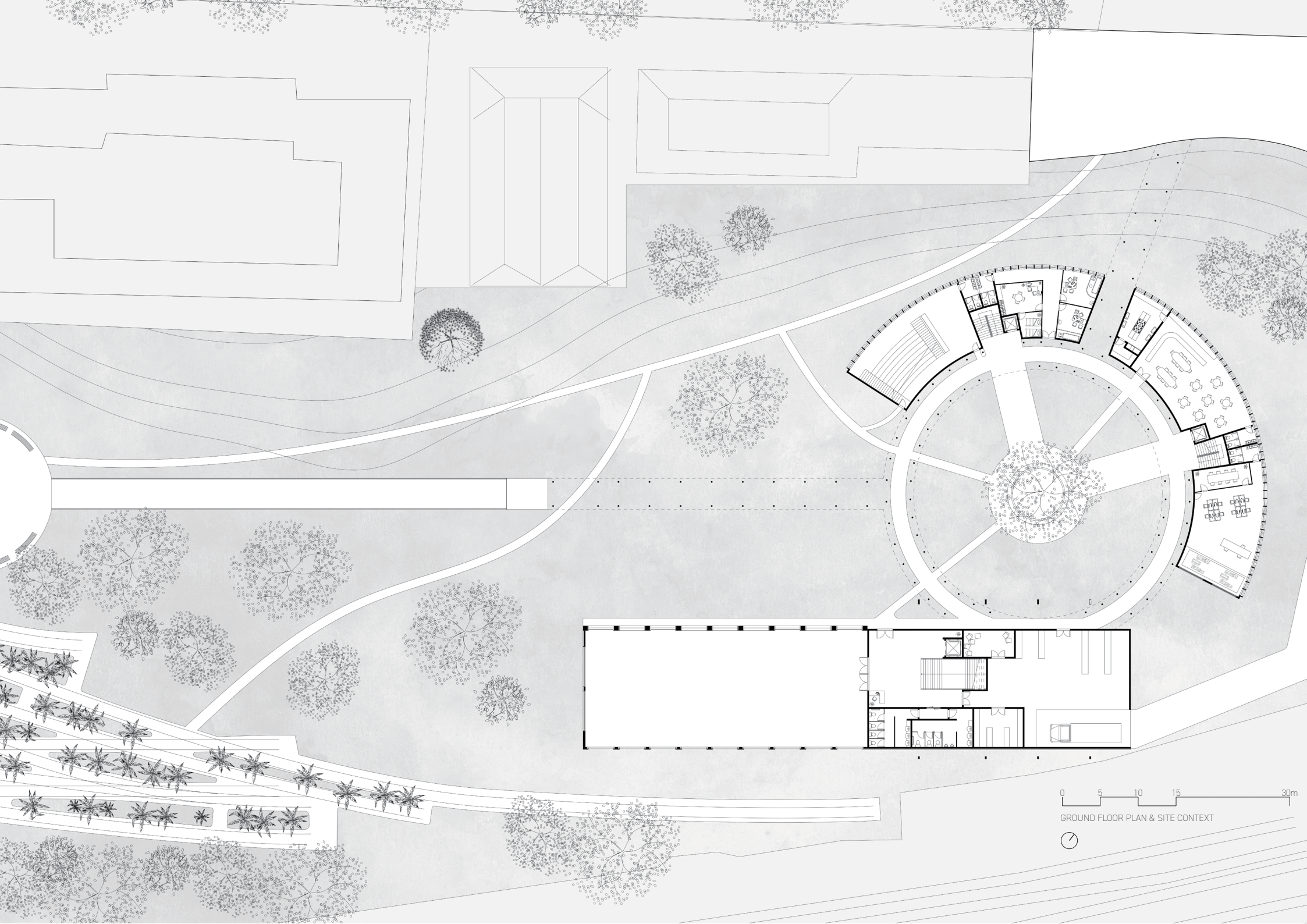
Conceived as a "tandem station" system, the layout emulates the linear logic of rail infrastructure to orchestrate public movement. Dual entry points channel pedestrian flows into a central tech hub, stitched together by a continuous, elevated walkway that translates the kinetic rhythm of a train journey into a multi-layered experience.

05 / VOLUMETRIC GRADUATION & SIGHTLINES

The massing responds directly to the active rail corridor through a tri-partite composition that steps upward from left to right. This graduated profile minimizes spatial oppression at entry points while securing critical view corridors, ensuring an unobstructed visual connection between the public domain and passing trains.

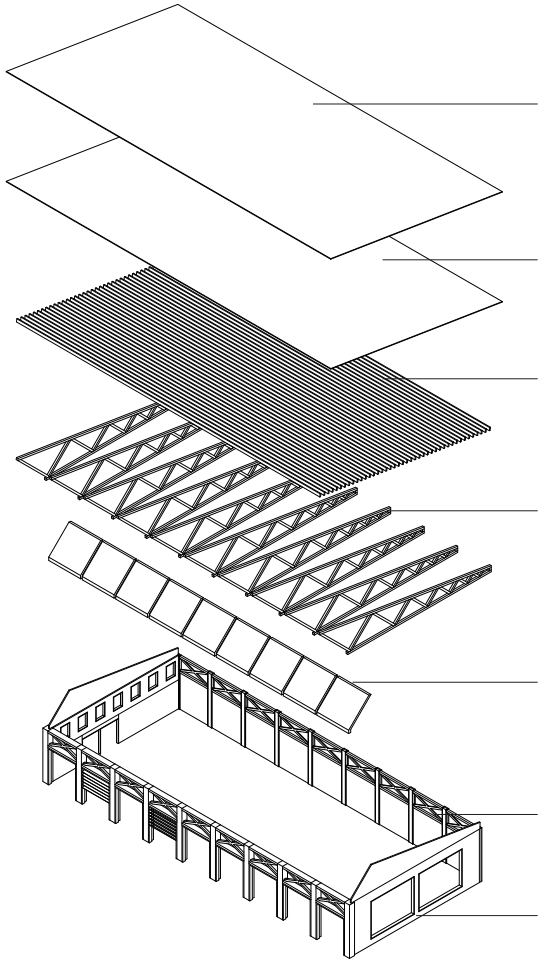


FIRST FLOOR PLAN & SITE CONTEXT



GROUND FLOOR PLAN & SITE CONTEXT

EXPLODED AXONOMETRIC



Weatherproof Membrane:
High-performance waterproofing layer protecting the structural timber substrate.

COR-TEN Steel Roof Cladding:
Weathering steel panels providing resilient environmental protection while establishing a unified material envelope.

Structural Timber Battens:
Secondary framing members providing rigid anchoring and ventilation gaps for the roof assembly.

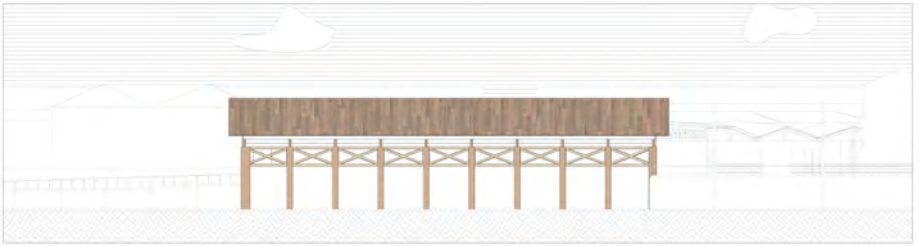
Truss:
A truss is an assembly of members such as beams, connected by nodes, that creates a rigid structure. This truss was created by warren truss deformation.

Glass window assembly:
Integrated glass ribbon designed to maximize diffused natural daylighting deep within the public hall.

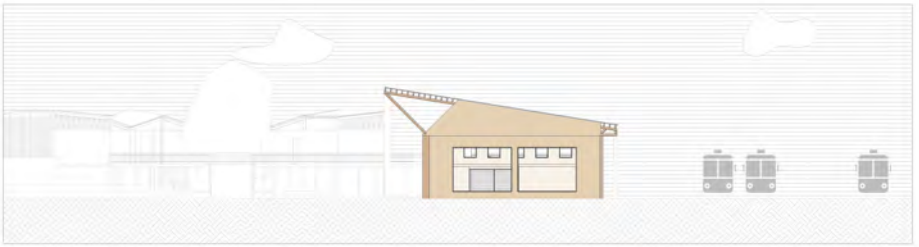
Mass Timber Frame:
Heavy timber column-and-beam system acting as the primary load-bearing structural skeleton.

Brick wall:
A robust masonry foundation wall that anchors the timber frame, balancing structural forces

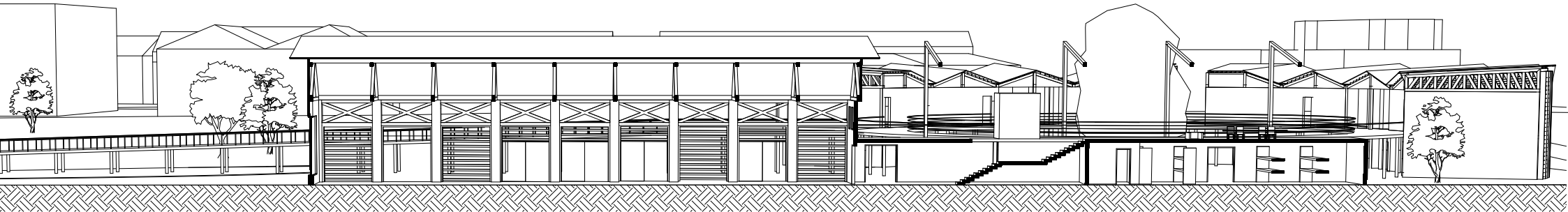
SOUTH ELEVATION



WEST ELEVATION



LONG SECTION



03 Gibbs Courtyard House

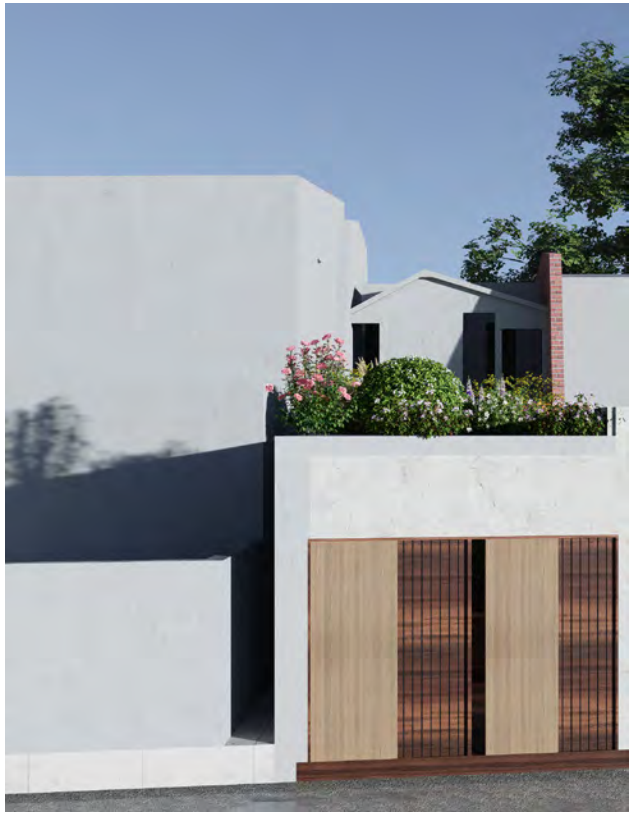
Location	25 Gibbes Street, Newtown, NSW 2042
Main Function	Low-rise Residential
Structure	Double Brick (Ex.), Timber Framing (New)
Tutor	JOHN POOLE
Year	2023
Key Words	Retrofit, Landscape Design, Green Roof, Courtyard
Software	Rhino, Illustrator, Photoshop

Development Application

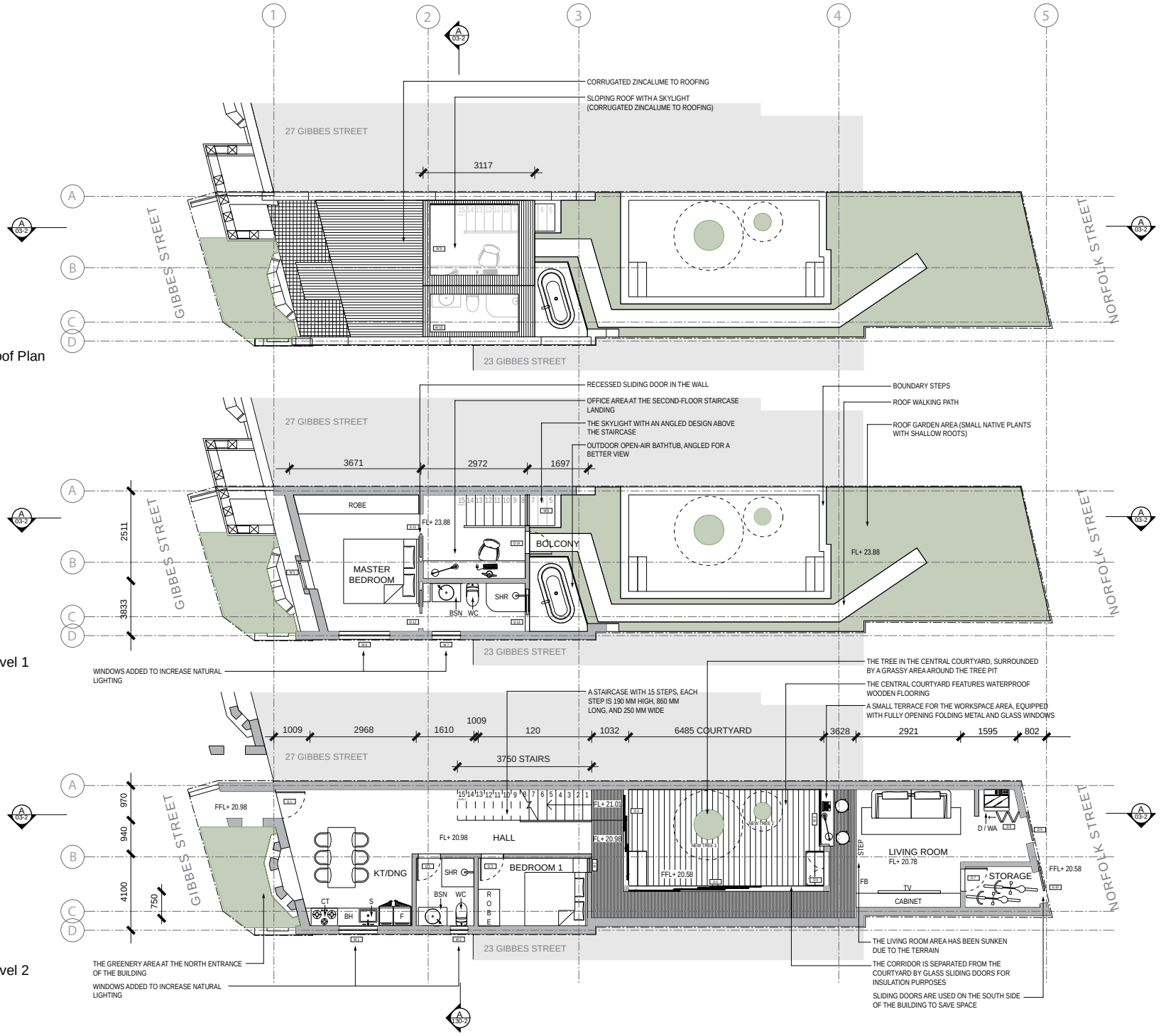
Alterations and Additions to
25 Gibbes Street, Newtown
Lot 1 DP 209329
Council Inner West Council

BASIX Commitments

Certificate No Pending / Draft
Hot Water System to be highly-energy efficiency electric heat pump
Rainwater Tank of at least 2000L, collecting runoff from min 20.32m² roof area
Cooling & Heating System to be 1-phase ducted airconditioning
Artificial Lighting to have a minimum of 80% LED lamps
Kitchen & Bathrooms to have individual exhaust fans with manual switch
Kitchen to install a gas cooktop and an electric oven
W1, North (Dining), timber framed, double glazing
W2, North (Level 2 Master Bed), sliding glass door, double glazing
W3, South (Level 2 Workspace), fully opening folding metal & glass window
SD1, South (Ground Living), sliding folding glass doors connecting to courtyard
Skylight 1, Above staircase, transparent roof board for natural light
Skylight 2, Above master bedroom, open skylight roof







1 Proposed Roof Plan
1:150

2 Proposed Level 1
1:150

3 Proposed Level 2
1:150

NOTE

ALL BUILDING WORK TO COMPLY WITH BUILDING CODE OF AUSTRALIA. RELEVANT AUSTRALIAN STANDARDS AND REVEVANT AUTHORITIES REGULATIONS.

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BH	BENCH
BK	BRICK/ BLOCKWORK - RENDERED
CON	CONCRETE
CT	COOKTOP
D	DRYER
DK	DECKING
DOO	DOOR
DP	DOWNPIPE
F	FRIDGE
FB	FLOORBOARDS
FBE	FLOORBOARDS (EXISTING)
FFL	FINISHED FLOOR LEVEL
G	GLAZING
GU	GUTTER
HWU	HOT WATER UNIT
J	JOINERY
LNY	LAUNDRY
M	MIRROR
MB	ROOF MEMBRANE ON PLYWOOD
R	ROBE
RC	REINFORCED CONCRETE
RL	RELATIVE LEVEL
S	SINK
SHR	SHOWER
SK	SKYLIGHT
ST	STEEL CLADDING
TB	TIMBER
TV	TELEVISION
WC	TOILET
WA	WASHING MACHINE
W	WINDOW

2023 TIANYU CAO

SITE:
25 GIBBES STREET
NEWTOWN

TITLE:
PROPOSED PLANS

SCALE:
1:150 @A4

DATE:
10/10/2023

DRAWN:
TIANYU CAO

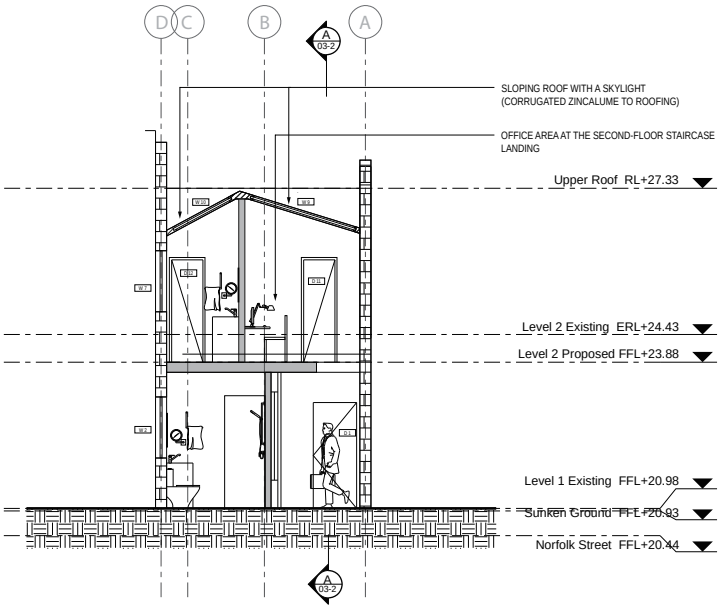
CHECKED:
JOHN POOLE

PROJECT NO:
C001

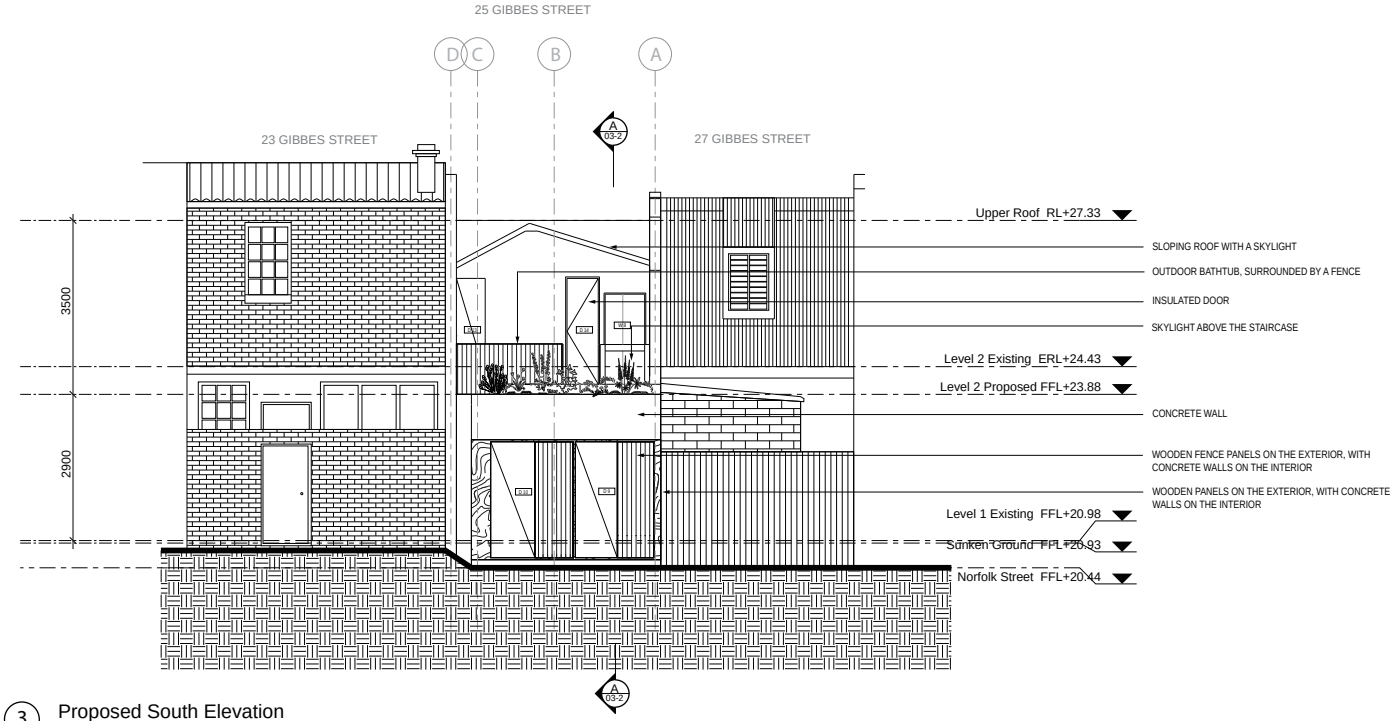
DRAWING NO:
A-01



1 Proposed North Elevation
1:150



2 Cross Section
1:150



3 Proposed South Elevation
1:150

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W	WINDOW

2023 TIANYU CAO

SITE:
25 GIBBES STREET
NEWTOWN

TITLE:
PROPOSED ELEVATIONS & SECTIONS

SCALE:
1:150 @A4

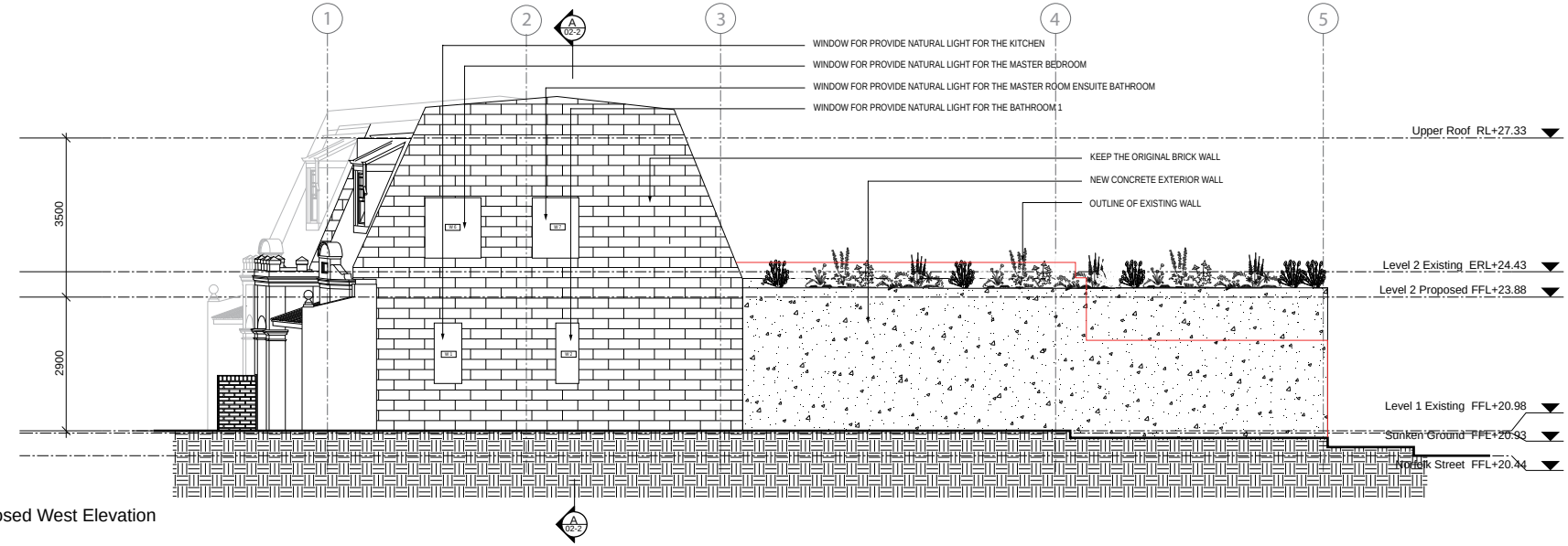
DATE:
10/10/2023

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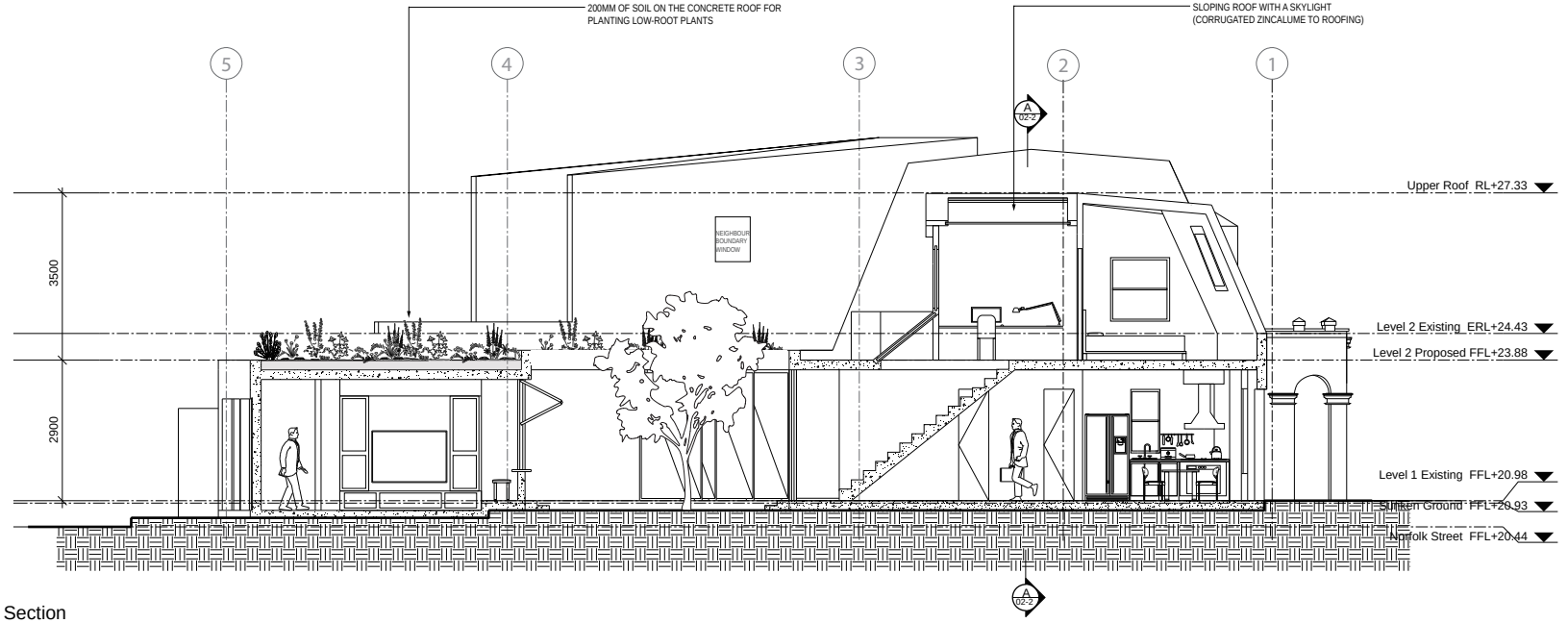
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JOHN POOLE

PROJECT NO:
C001

DRAWING NO:
A-02



1 Proposed West Elevation
1:150



2 Proposed East Section
1:150

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WC	TOILET
WA	WASHING MACHINE
W	WINDOW

2023 TIANYU CAO

SITE: 25 GIBBES STREET NEWTOWN

TITLE: TECHNICAL DETAILS & ELEVATIONS

SCALE: 1:150 @A4

DATE: 10/10/2023

DRAWN: TIANYU CAO

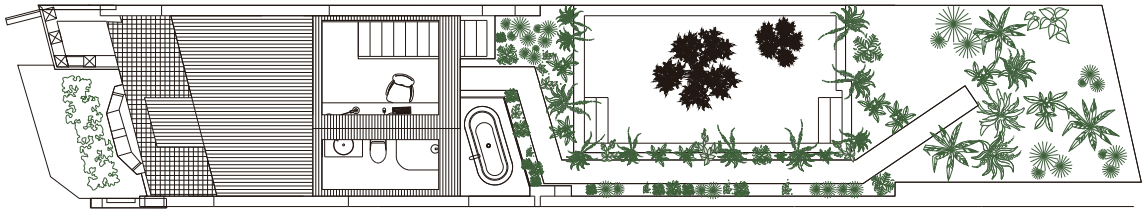
CHECKED: JOHN POOLE

PROJECT NO: C001

DRAWING NO: A-03

LANDSCAPE SCHEDULE

*This design positions the landscape as a natural extension of the architectural space, establishing rich layers of light, shadow, and visual depth through precise planting selection. **In the front courtyard**, evergreen Viburnum (over 1m) is utilized as a green screen, balancing low-maintenance requirements with an excellent sense of spatial enclosure. **Small shrubs are introduced on the first floor**, featuring drought-tolerant, drought-resilient native Australian and adaptive species that frame the pathways with a soft, textured boundary. **Within the central courtyard**, a mature, seasonal Japanese Maple serves as the spatial focal point, anchoring the visual core of the vertical elevation. This creates a sensory landscape narrative of touch, scent, and sight, allowing residents to experience the rhythmic cadence of the seasons amidst an urban context.*





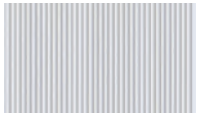
Planting Legend			
Symbol	Botanical Name	Common Name	Category
	Themeda australis	Kangaroo grass	Under 1m
	Callistemon viminalis	Crimson Bottlebrush	Under 1m
	Dianella caerulea	King Alfred	Under 1m
	Westringia eremicola	Rosemary officinalis prostrate	Under 1m
	Rhododendron	Rhododendron lochiae	Under 1m
	Clivia	Natal lily	Under 1m
	Viburnum odoratissimum	Sweet viburnum	Over 1m
	Lomandra	Katrinus	Under 1m
	Eremophila	Eremophila debilis	Under 1m
	Acer palmatum	Japanese Maple	4.5m-10m

MATERIALITY & ELEVATIONS

This layout presents the project's elevations and exterior material palette, demonstrating a modern residential intervention that respects the local heritage context. The street-facing facade retains and restores the original brick texture in compliance with heritage regulations, preserving historic continuity. The rear and internal extensions employ contemporary materials: corrugated Zincalume and raw concrete walls create sleek, industrial lines, while wooden cladding and timber doors add warmth to break up the cool-toned surfaces. Large reinforced glass skylights maximize vertical natural light, allowing sunlight to interact dynamically with the shifting textures of the facade.



PROPOSED MATERIALS



CORRUGATED ZINCALUME TO ROOFING



INSULATED WOODEN DOOR



ALUMINUM DOOR WITH GLASS



BRICK TO BE RENEWED IN THE SAME OR SIMILAR COLOR BRICK



CONCRETE WALL



WOODEN TEXTURED EXTERIOR CLADDING



CORRUGATED ZINCALUME ROOF WITH REINFORCED GLASS

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2023 TIANYU CAO

SITE: 25 GIBBES STREET NEWTOWN

TITLE: MATERIALITY & LANDSCAPE DESIGN

SCALE: 1:150 @A4

DATE: 10/10/2023

DRAWN: TIANYU CAO

CHECKED: JOHN POOLE

PROJECT NO: C001

DRAWING NO: A-04